Normalization Practice Questions

1. Museum inventory. A museum stores all restoration dates for an artifact in one cell, such as `2019-04-12, 2023-08-06`. The museum needs to query each restoration date separately. Which normal form should it address first?

a. 1NF · b. 2NF · c. 3NF · d. BCNF

2. Consider the following: `R(A, B, C, D) with B → CD, and A → C`. What normal form does `C → A` violate?

a. 1NF · b. 2NF · c. 3NF · d. BCNF

3. Bicycle rentals. In `Rental(RentalID, BikeID, StartTime, Duration)`, `RentalID` determines all other attributes. The combination `(BikeID, StartTime)` also determines all other attributes. Which option lists **all** candidate keys?

a. `RentalID` only
b. `(BikeID, StartTime)` only
c. `RentalID` and `(BikeID, StartTime)`
d. `RentalID` and `BikeID`

4. Veterinary clinic. In `Visit(VisitID, PetID, VisitDate, MicrochipID)`, `VisitID` determines the other attributes, and `PetID → MicrochipID`. What is the **highest** normal form the table satisfies?

a. 1NF · b. 2NF · c. 3NF · d. BCNF

5. Radio broadcasting. `Broadcast(StationID, ShowID, ProducerID, Airtime)` has exactly two candidate keys, `(StationID, ShowID)` and `(StationID, ProducerID)`. Each producer works on one show, giving `ProducerID → ShowID`. Why does that dependency violate **BCNF** while satisfying the **3NF** rule?

a. `ProducerID` is a superkey, and `ShowID` is non-prime.
b. `ProducerID` is not a superkey, and `ShowID` is prime.
c. `ProducerID` is not a superkey, and `ShowID` is non-prime.
d. `ProducerID → ShowID` is trivial.

6. Package tracking. In `Parcel(TrackingNumber, LabelBarcode, Weight)`, `TrackingNumber` and `LabelBarcode` each uniquely identify a parcel. These are the only determinants of non-trivial dependencies with minimal left sides. What is the **highest** normal form the table satisfies?

a. 1NF · b. 2NF · c. 3NF · d. BCNF

7. In `R(A, B, C, D)`, `A → B` and `B → CD`. What is the highest normal form the table satisfies?

a. 1NF · b. 2NF · c. 3NF · d. BCNF